

Prime Factors and Powers

Prime Factors: Definition

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A **prime number** is a number with exactly two factors: 1 and itself.

Examples:

- 2, 3, 5, 7, 11, 13, ...

A **prime factorisation** writes a number as a product of prime numbers.

Example: Prime Factorisation

Write 84 as a product of prime factors.

$$\begin{aligned}84 &= 2 \times 42 \\&= 2 \times 2 \times 21 \\&= 2^2 \times 3 \times 7\end{aligned}$$

Calculator Check

You should always check your prime factorisation using a calculator.

Most scientific calculators have a **FACT** function which lists the prime factors of a number.

The prime factors must multiply back to the original number.

Prime Factorisation with Algebra

We can also factorise algebraic expressions.

$$7a^2b = 7 \times a^2 \times b$$

$$49ab^3 = 7^2 \times a \times b^3$$

This will be useful when finding HCF and LCM of algebraic terms.

Highest Common Factor (HCF)

The **HCF** is the largest number that divides exactly into both numbers.

Method:

- 1) Write each number as a product of prime factors.
- 2) Write down **who loses** for each prime factor.

i Note

None of a prime factor loses against one or more of a prime factor, e.g. no 7s loses to 7.

Example: HCF

Find the HCF of 84 and 360.

$$84 = 2^2 \times 3 \times 7$$

$$360 = 2^3 \times 3^2 \times 5$$

Common factors:

$$2^2 \times 3$$

$$\text{HCF} = 12$$

Lowest Common Multiple (LCM)

The **LCM** is the smallest number that both numbers divide into exactly.

Method:

- 1) Write each number as a product of prime factors.
- 2) Write down **who wins** for each prime factor.

Example: LCM

Using the same factorisations:

$$84 = 2^2 \times 3 \times 7$$

$$360 = 2^3 \times 3^2 \times 5$$

$$\text{LCM} = 2^3 \times 3^2 \times 5 \times 7 = 2520$$

HCF and LCM Check

For two positive integers a and b :

$$\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$$

Here:

$$12 \times 2520 = 84 \times 360$$

This confirms the answers are correct.

Laws of Indices

For $a \neq 0$:

$$a^m \times a^n = a^{m+n}$$

$$a^m \div a^n = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

$$(ab)^n = a^n b^n$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

Example: Using the Laws of Indices

Simplify:

$$(3x^2y^5)^3$$

$$\begin{aligned}(3x^2y^5)^3 &= 3^3(x^2)^3(y^5)^3 \\ &= 27x^6y^{15}\end{aligned}$$

Powers of Zero

Consider:

$$\frac{5^3}{5^3}$$

Using the laws of indices:

$$5^{3-3} = 5^0$$

But:

$$\frac{5^3}{5^3} = 1$$

So:

$$a^0 = 1 \quad (a \neq 0)$$

Negative Indices

Using division:

$$3^5 \div 3^7 = 3^{-2}$$

But:

$$\frac{3^5}{3^7} = \frac{1}{3^2}$$

So:

$$a^{-n} = \frac{1}{a^n}$$

Common Pitfall with Negative Indices

Compare:

$$\frac{1}{2x} = \frac{1}{2}x^{-1}$$

and

$$\frac{2}{x} = 2x^{-1}$$

Only the **variable** is affected by the negative index, not the coefficient.

Fractional Indices

Fractional indices represent roots.

$$a^{\frac{1}{n}} = \sqrt[n]{a}$$

$$a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

Example:

$$x^{\frac{2}{3}} = \sqrt[3]{x^2}$$

Example: Fractional Indices

Simplify:

$$c^{\frac{1}{2}} \left(c^{\frac{3}{2}} - c^{\frac{1}{2}} + c^{-\frac{1}{2}} \right)$$

$$= c^2 - c + 1$$

Test Your Understanding

- 1) Write 540 as a product of prime factors.
- 2) Find the HCF of 72 and 90.
- 3) Find the LCM of 24 and 60.
- 4) Simplify:

$$(2p^3q^{-2})^2$$

- 5) Simplify:

$$y^{-\frac{1}{2}} \times y^{\frac{3}{2}}$$

Answers

1) $540 = 2^2 \times 3^3 \times 5$

2) $\text{HCF} = 18$

3) $\text{LCM} = 120$

4) $4p^6q^{-4} = \frac{4p^6}{q^4}$

5) y